

Alexander Belik

Software Engineering | AI Systems | Scientific Computing

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PROFILE

Theoretical physics graduate and current M.Sc. High-Performance Computing student at Trinity College Dublin. I build research platforms, GPU and HPC simulation tooling, machine-learning systems, and full-stack products across theoretical physics, scientific computing, and data-heavy software.

EDUCATION

M.Sc. High-Performance Computing

Trinity College Dublin, current

B.A. Theoretical Physics

Trinity College Dublin, 2022-2026

EXPERIENCE

Scientific Software Developer

Trinity College Dublin

May 2026 - Sep 2026

- Built a digital lab notebook platform storing experimental measurements as structured, auditable records.
- Developed GPU-accelerated physics demonstrations that run headlessly on the Leonardo EuroHPC system and on Brain++ Discoverer GB200 nodes, for the EuroHPC Demo Lab 2026.

Teaching Assistant

Trinity College Dublin

Oct 2025 - May 2026

- Led weekly tutorials and problem-solving sessions aligned with lectures.
- Graded assignments and exams with clear, constructive feedback.

Research Assistant

Trinity College Dublin, CRANN Institute

Jun 2025 - Sep 2025

- Designed Echo State Network reservoir models for quantum time-evolution data.
- Analysed model stability, convergence, and predictive accuracy across hyperparameters.

Software Engineer (EdTech)

Grinds360

Jun 2024 - Dec 2025

- Built Python automation tools for video and interactive lesson workflows.
- Contributed to UX iteration in Figma and full-stack product features.

Experimental / Computational Research Assistant

Trinity College Dublin

May 2024 - Sep 2024

- Created a Python spectroscopy control system for ferrofluids and ferronematic liquid crystals.
- Built automated acquisition and analysis workflows for repeated experimental data.
- Contributed to liquid-crystal research that led to co-authored publication work.

Exam Corrector

The Dublin Academy of Education

Jul 2021 - Jan 2024

- Corrected exams and coursework and supported marking workflows.

Selected work

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HOME PLATFORM

Inhabis (in progress)

A local-first operating layer for a connected home: one trusted model of the house, explicit policy, and evidence that an action actually worked.

- TypeScript server with separated layers for operational state, derived planning context, and constrained planning
- Embedded Matter controller with one persistent fabric per home, plus an Android companion that does BLE and Thread onboarding next to the device while the server finishes commissioning over the LAN

Tools: TypeScript, Node.js, React, Python, Matter / Thread, Kotlin

DESKTOP WORKSPACE

Orion Mini (in progress)

Local-first Windows workspace for projects, tasks, reminders, transcription, and an assistant, with React rendering inside a Qt shell.

- Multi-track transcription library with checkpoints, corrections, anchored notes, and timestamped exports
- Activity-aware reminders with idle detection and escalating native alerts that cannot be quietly dismissed

Tools: Python, PySide6 / Qt, React, TypeScript, CUDA, Whisper

RESEARCH DATA PLATFORM

LabFlow

Digital lab notebook for a materials research institute that stores measurements as structured relational records.

- React and TypeScript frontend over a FastAPI and PostgreSQL backend with versioned migrations
- Sample, project, instrument, and booking models with role-based access across organisations and sites

Tools: React, TypeScript, FastAPI, PostgreSQL, Docker, Tailwind CSS

HPC AND GPU SIMULATION

EuroHPC Demo Lab

Fourteen GPU physics and AI demonstrations for the EuroHPC Demo Lab 2026, rendered on Leonardo and Discoverer and replayed on a public stand.

- Fourteen solvers spanning lattice-Boltzmann flow, particle-mesh cosmology, direct N-body gravity, black-hole lensing, reaction-diffusion, plasma control, molecular dynamics, and two evolving-AI games
- One demo contract that runs on NumPy, CuPy/CUDA, or a hybrid pipeline overlapping GPU solving with CPU frame encoding

Tools: Python, CuPy / CUDA, PyTorch, NumPy, SLURM, JavaScript

[View on GitHub](#)

LOCAL AI ASSISTANT

ORION (in progress)

Windows-first local assistant with wake-word gating, local LLM orchestration, memory, guarded tools, and a PyQt overlay/dashboard.

- LLM-first capability selection with typed tool boundaries
- Wake-word and follow-up metadata for safer utterance handling

Tools: Python, PyQt, Ollama, SQLite, Whisper, Local LLMs